

Question 1

Part 1.

```
N=256; M=400; %% Over-sample for error checking
L = pi; d=1; d2=d*d;
```

The above lines set the number of Fourier modes and Legendre points, as well as the domain length and operator coefficients on u and u'' .

```
[Ah,Bh,Ch,Dh,z,w]=semhat(M);
Ab = (2/L)*Ah; Bb = (L/2)*Bh; Ib = eye(M+1); R = Ib(2:end-1,:);
A=R*Ab*R'; B=R*Bb*R'; H = d2*A + B;
```

The above lines call `semhat`, a function which computes the stiffness, mass and convection matrices for a 1D single Spectral Element. In addition, `semhat` returns the GLL points and weights through a call to `zwgll`. R is a restriction matrix to apply homogenous Dirichlet boundary conditions.

The resulting A and B matrices are the stiffness and mass matrices, which are constructed into H , the Helmholtz matrix.

```
z=L*(1+z)/2; w=L*w/2; % Scale points and weights to [0,L]
xb=z; fb=1+0*xb; % Set RHS for Legendre spectral soln
us= R'(H\ (R*Bb*fb)); % Solve Legendre system
```

These lines of code set up the forcing function for the Legendre spectral solution. Then, the Helmholtz problem is solved for the coefficients.

```
plot(xb,us,'k-', 'linewidth',1.4); hold on;
```

Plots Legendre solution.

```
uf=0*xb; %% Initialize Fourier solution on GLL pts
for k=1:2:N; k1=k+1;
    sk = sin(k*xb);
```

Fourier Sin series is used, looking for solution on the same points that we used for Legendre so we can compare.

```
% fk = 2*(w'*sk)/(2*pi); % 2 in numerator because integrate over [-pi,pi]
fk = 2/(k*pi); % 2 in numerator because integrate over [-pi,pi]
```

The fourier coefficients for a square wave are pre-computed here.

```
uk = 2*fk./(1+d2*k*k); % 2 in numerator because +/- k mode.
uf = uf + uk*sk;
```

Add contribution from the kth component of the fourier series to the overall values.

```
ef=uf-us;  
em(k1/2)=max(abs(ef));
```

Compute error compared to Legendre solution (L_{∞})

```
ef2(k1/2) = sqrt(ef'*Bb*ef/pi);
```

Energy norm of error (L_2)

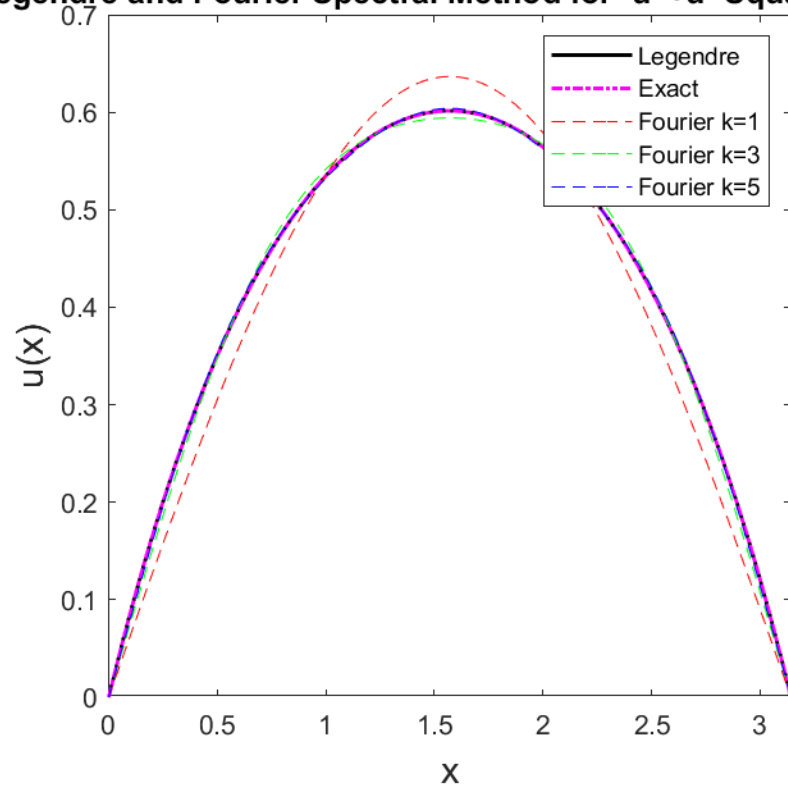
```
nN(k1/2) = k;  
if k==1; plot(xb,uf,'r--'); end;  
if k==3; plot(xb,uf,'g--'); end;  
if k==5; plot(xb,uf,'b--'); end;
```

```
end;
```

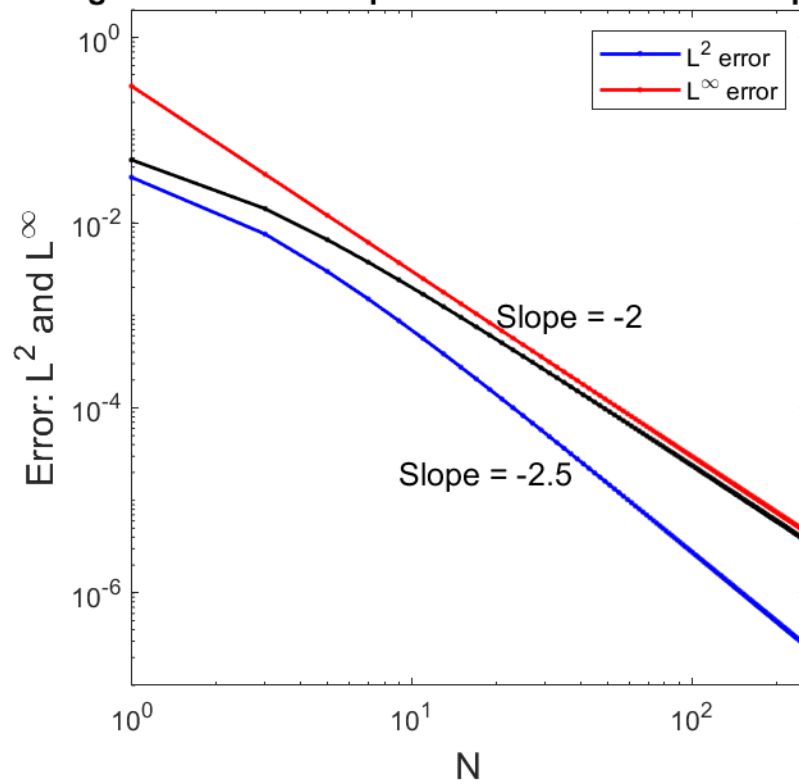
The above lines and the rest of the code is just for plotting purposes.

I ran this code and also added the exact solution to the plot, just to make sure that it lined up with the M=400 Legendre.

Legendre and Fourier Spectral Method for $-u''+u=\text{Square Wave}$



Convergence of Fourier Spectral Method for $-u''+u=\text{Square Wave}$



Part 2.

Consider the problem

$$-u'' + u = \pi x - x^2$$

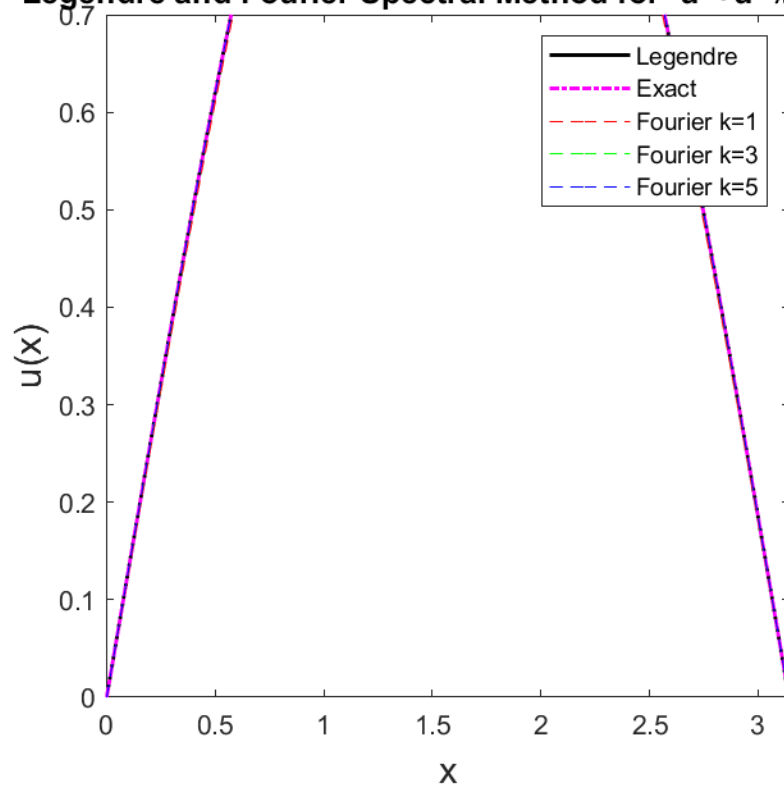
The odd Fourier coefficients f_k are then:

$$f_k = \frac{8}{\pi k^3}$$

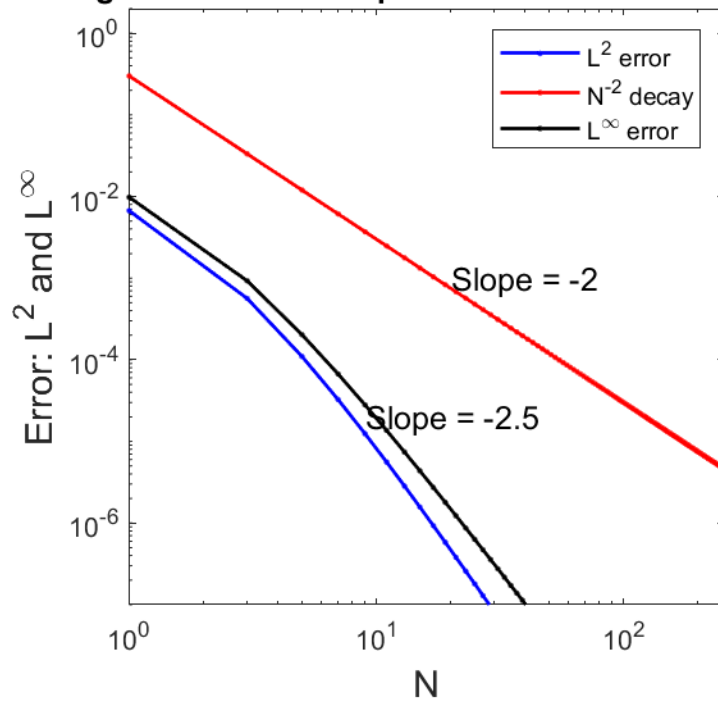
(ChatGPT figured out these coefficients).

The corresponding plots are:

Legendre and Fourier Spectral Method for $-u''+u=\pi x -x^2$



Convergence of Fourier Spectral Method for $-u''+u=\pi x -x^2$



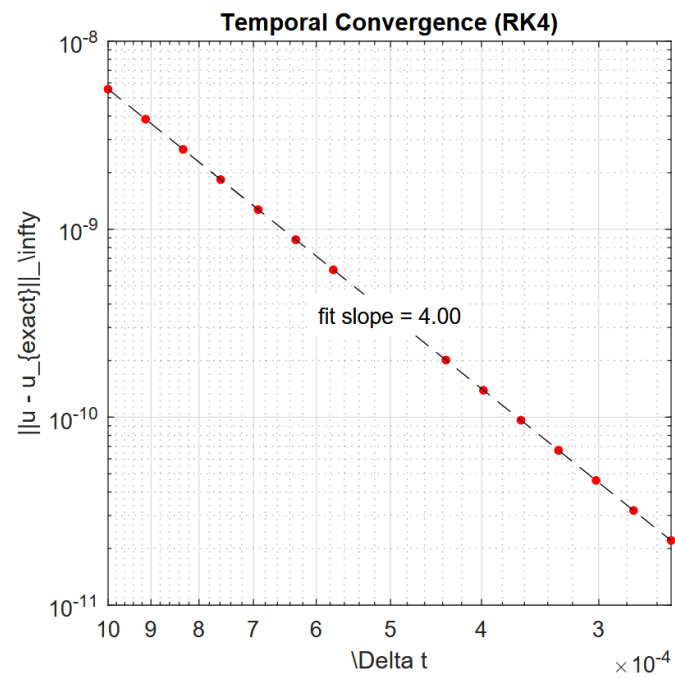
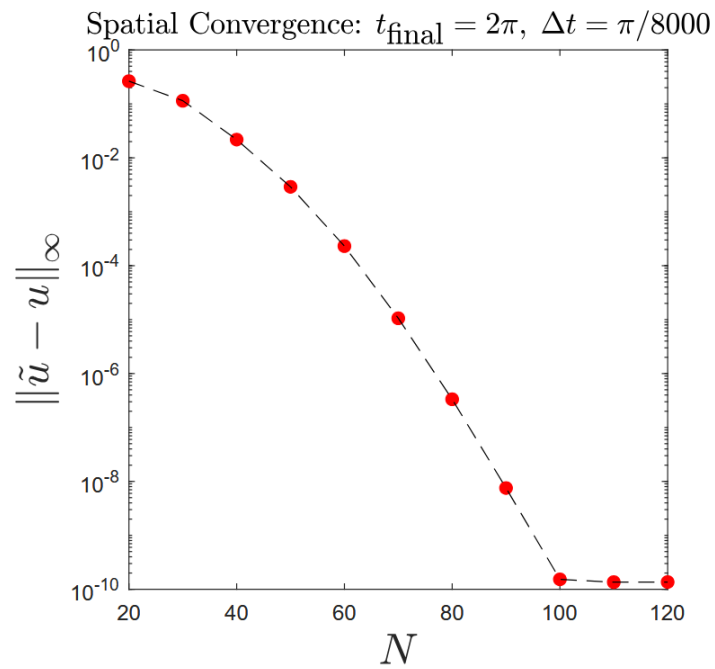
These demonstrate better than power-law convergence (ie exponential convergence) which is expected for a smooth function.

Question 2.

Part a.

Flowchart documenting ck4_0.m code process submitted in separate .pdf file.

Part b.



The first plot clearly shows exponential convergence, and the second shows power-law convergence of order 4. Of note in the first plot is the error threshold which is reached after $N=100$. This is the threshold at which the temporal error dominates for the given timestep of $\pi/4000$.

To capture the 4th-order temporal convergence behavior, the convergence comparison must be made at sufficient N such that the temporal error dominates. If Δt was allowed to further decrease, a threshold would be reached where the spatial error dominated and the 4th order temporal convergence would stop.

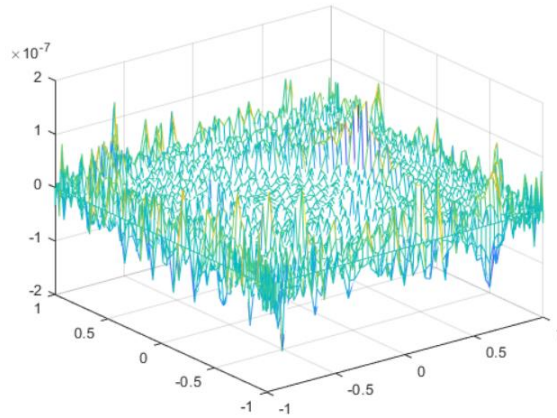


Figure: Error when spatial error dominates

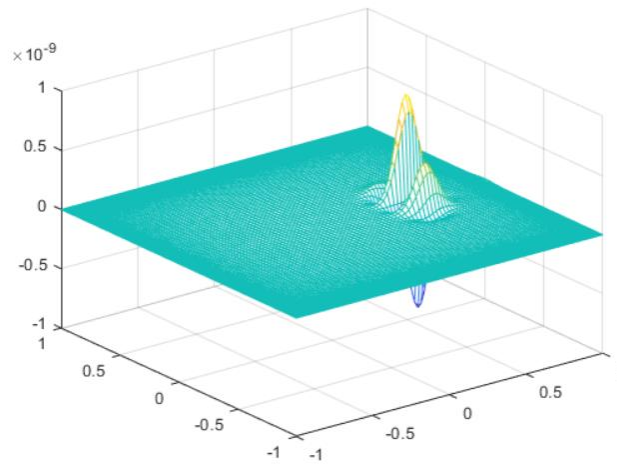


Figure: Error when temporal error dominates

When temporal error dominates, the error appears in a coherent structure associated with the wave form moving around the domain. When spatial error dominates, the error appears in noise distributed mostly randomly around the domain.

Other errors might be associated with domain boundary conditions. If there is any interaction between the waveform and the boundary, some energy may be spuriously reflected back into the domain. I think some of this error is present at roughly $x=1$, $y=0.5$ in the spatial error plot.

The temporal error looks like it is associated with diffusion error on the waveform. It does not appear there is any dispersion error.

There is no iterative solver error, because the problem is small enough that the matrix is solved directly. However, there may be error from the direct matrix solution.